Giovanni Madrigal, PhD

817–443-1487 | giovanni.madrigal12@yahoo.com | LinkedIn | Github | Website | Google Scholar

Experience

Postdoctoral Researcher Sep. 2026 – Present
University of Kansas *Lawrence, KS*

- Applying whole-genome approaches to characterize mito-nuclear interactions in Evolutionary Mutant Models as a framework for understanding human mitochondrial disease

Staff Scientist May 2025 – May 2026
Frenzy Crop Systems Inc *Champaign, IL*

- Directed genomic marker discovery efforts for agronomically important traits in major crop species, leveraging whole-genome sequencing data
- Served as sole cloud administrator for company AWS environment, establishing and maintaining EC2 compute instances, S3 storage for genomic data, and IAM user provisioning
- Developed custom bioinformatic tools and pipelines in Python, C++, and R to address analytical needs

Graduate Research Assistant May 2021 – Aug. 2026
University of Illinois Urbana-Champaign *Champaign, IL*

- Developed Klumpy, a Python-based tool for evaluating long-read genome assembly integrity, featuring novel algorithms for detecting sequence motifs with minimal dependencies
- Developed Synolog, a scalable C++ framework for synteny-based genome architecture characterization, incorporating custom visualization modules and algorithms designed for large-scale comparative genomics
- Identified a candidate gene underlying phenotypic variation through population-level genomic marker discovery
- Collaborated with 8 research groups across multiple departments and universities on projects spanning single-cell RNA-seq, eQTL mapping, genome assembly, and population genomics

Laboratory Technician Jan. 2021 – May 2021
University of Miami Rosenstiel School of Marine and Atmospheric Science *Miami, FL*

- Analyzed population structure and genetic diversity of an endangered species to support conservation management

Technical Skills

Languages: Python, C++, Bash, R, SQL, HTML/CSS
Developer Tools: Git, Conda, GDB
Cloud & HPC: Amazon Web Services (EC2, S3, IAM), Slurm
Computational Methods: Parallelization, data visualization, algorithm design
Genomics & Transcriptomics: Genome assembly, marker discovery, Single-cell and bulk RNA-Seq analysis

Publications and Manuscripts

Software

Madrigal, G., & Catchen, J. M. (2026). *Synolog: A scalable synteny-based framework for genome architecture characterization* [Under Review].

Madrigal, G., Minhas, B. F., & Catchen, J. (2025). Klumpy: A tool to evaluate the integrity of long-read genome assemblies and illusive sequence motifs. *Molecular ecology resources*, *25*(1), e13982.

Applied Genomics

Benjamin, N. R., **Madrigal**, G., Ishida, Y., Catchen, J., Allen, K. L., Pepin, B., & Roca, A. L. (2026). Genomic validation of perv-c-free pigs to support xenotransplantation. *Xenotransplantation*, *33*(1), e70109.

Education

University of Illinois Urbana-Champaign Urbana, IL
Ph.D. in Biology *May 2021 – Aug. 2026*

Texas A&M University at Galveston Galveston, TX
Master of Science in Marine Biology *Jan. 2018 – Dec. 2020*

Texas A&M University at Galveston Galveston, TX
Bachelor of Science in Marine Biology and Marine Fisheries *Aug. 2013 – Dec. 2017*